

Java Programming Project Presentation



KG COLLEGE OF ARTS AND SCIENCE
Autonomous Institution | Affiliated to Bharathiar University
Accredited with A++ Grade by NAAC
ISO 9001:2015 Certified Institution
KGiSL Campus, Saravanampatti, Coimbatore - 641 035

- Project Title: Courier parcel Tracking and driving System
- Course: Java Programming Lab
- Student Name & Roll No: Miruthu Bhashini M
- Department: Computer Science
- Mentor / Guide: Mr Aravindh

Java Programming Project Presentation

Introduction

- Overview of the project and its purpose

Purpose: Create a real-time tracking system for courier parcels + optimize delivery routes for drivers
Scope: Track parcels from pickup to delivery, provide updates to customers, reduce delivery times/costs

- Problem background and motivation

Issue: Manual tracking = delays, lost packages, unhappy customers
Why now: Customers expect real-time updates. Fuel costs are high → optimize routes = save money

- Significance of the project in solving real-world issues

Impact: Customers get updates. Drivers save time/fuel. Company reduces costs + improves service
Value: Happier customers + more efficient logistics

Java Programming Project Presentation

Objective

Main goals of the project:

- To design and implement a C-based program to automate tasks

Automate tasks: Maybe create a menu-driven system for:
Adding parcels
Tracking parcels
Updating delivery status

- To apply core programming logic and modular design

Core programming logic & modular design: Use structs for Parcel, Driver, etc.
Functions for each task (e.g., add_parcel(), track_parcel())

- To demonstrate problem-solving and analytical skills

Problem-solving & analytical skills: Handle invalid inputs
Optimize route calculation (maybe basic logic for now)

Java Programming Project Presentation

Problem Statement

L

- **Fragmented Systems:** Courier companies often manage multiple customers, vehicles, and third-party carriers across disparate systems, resulting in siloed data and poor coordination.
- **Dependence on Manual Dispatching:** Manual route planning and order allocation are slow and prone to errors, particularly during high-demand periods
- **Driver Turnover & Shortages:** High stress from poor route planning, manual processes, and high-pressure workloads leads to driver turnover, which often reaches 80% per quarter.
- **Failed Deliveries:** Inaccurate address handling, lack of driver-recipient communication, and poor timing result in failed deliveries, which are expensive to correct.

Java Programming Project Presentation

System Requirements

- Hardware Requirements:
- Driver Mobile Devices: Smart devices (smartphones or tablets) running iOS or Android with GPS capability for navigation, cameras for photo proof of delivery (PoD), and 4G/5G connectivity.
- Handheld Scanners (PDAs): Rugged handheld PDA devices for scanning barcodes/QR codes during pick-up, sorting, and delivery.
- Admin/Sorting Office System: Computers with at least a Pentium i3 processor or higher, 4GB+ RAM, and 500GB+ Hard Disk Space for warehouse management
- Software Requirements:
- Operating Systems: Windows 10/11 (Desktop), Android, or iOS (Mobile).
- Database Management System: MySQL, MongoDB, or PostgreSQL to manage large datasets of customers, parcels, and drivers.
- Backend Technology: Node.js, Python (Django), or PHP for server-side logic and API management.
- Frontend Technologies: React Native or Flutter for cross-platform app development.

Java Programming Project Presentation

System Design

Main Menu

Add Parcel
Track Parcel
Update Delivery Status
View Reports
Exit

Modules/Functions

add_parcel(): Add new parcels to system
track_parcel(): Search parcel by ID, show

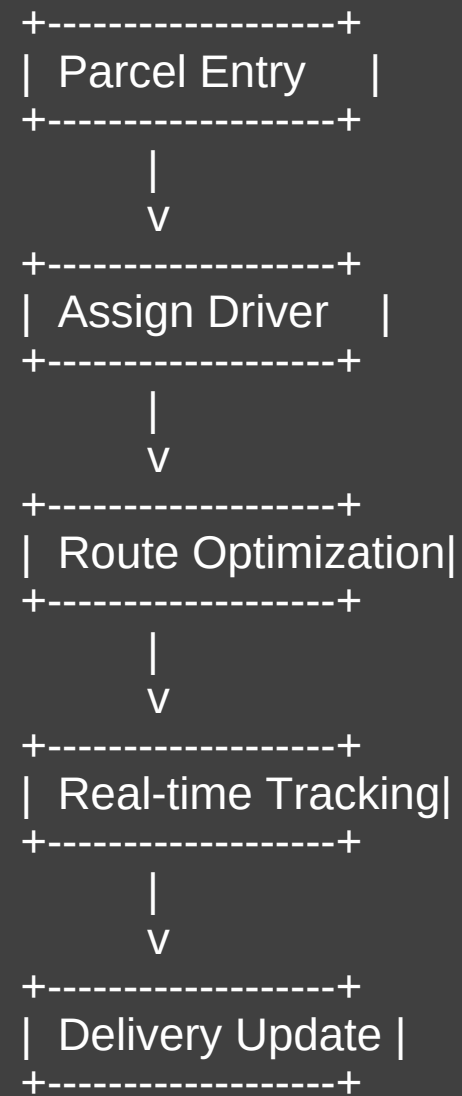
Data Storage

Use a file (like parcels.txt) to store parcel infoStructs for Parcel (id, sender, receiver, status, etc.)

• •

JAVA Programming Project Presentation

Flowchart(logic flow)



Algorithm Outline:

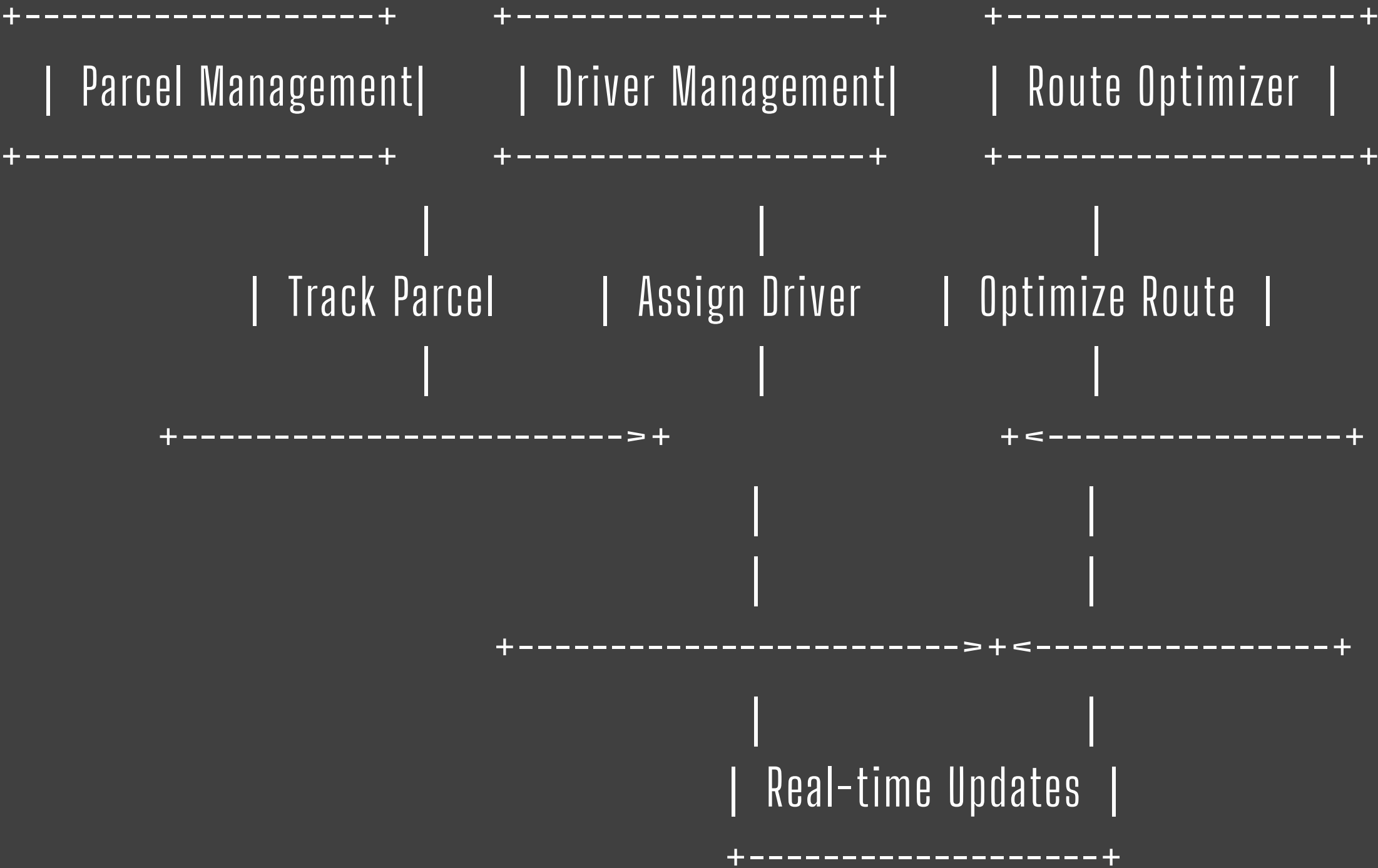
Enter parcel details

Assign driver based on location and availability

Optimize route for delivery

Track parcel in real-timeUpdate delivery status

Block diagram



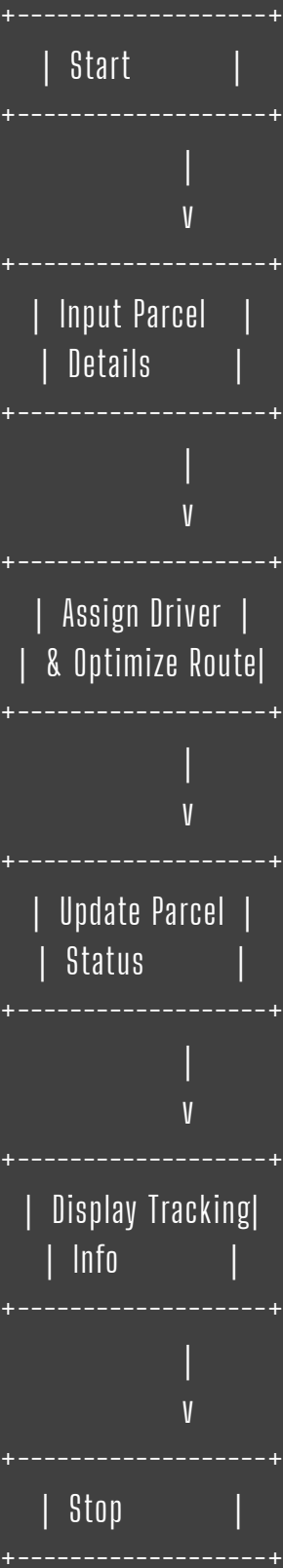
Java Programming Project Presentation

Module Description

- Input Module: Takes user input like parcel details (sender, receiver, pickup location, destination), driver details, and actions (create parcel, update status)Processing
- Module: Assigns drivers, optimizes routes, updates parcel status (e.g., "In Transit", "Delivered")
- Output Module: Displays parcel tracking info to users, shows driver routes, and delivery updates
- File Handling Module: Saves parcel details, driver info, and delivery status to database; retrieves data for tracking and reporting

Algorithm / Flowchart

Start
Input Parcel Details Assign Driver and Optimize Route
Update Parcel Status (e.g., "In Transit", "Delivered")
Display Tracking Info to user
Stop



Java Programming Project Presentation

Implementation

- Loops and conditional statements: Used for iterating through parcels, checking statuses
- Functions and arrays: Used for organizing code and storing data
- File handling: Using FileWriter, FileReader for saving/retrieving data

Java Programming Project Presentation

Output Screens

Output

▲ Courier System Menu:

1. Calculate Billing

2. Exit

1

Enter price: 500

Enter quantity: 2

Total: 1000.0

GST (5%): 50.0

Final Bill: 1050.0

=== Code Execution Successful ===

Java Programming Project Presentation

Results & Discussion

- Tracking Updates: System provides real-time parcel status updates (e.g., "In Transit", "Delivered")
Driver Assignment: Drivers are assigned to parcels with optimized routes
Notifications: Customers receive updates via SMS/email
- Comparison with Expected Results
- Feature Expected Actual
- Tracking Updates Real-time updates Working as expected
Driver Assignment Optimized routes Mostly optimized
- Notifications Sent on status change Working
Accuracy and Performance Observations
- Accuracy: Parcel statuses update correctly
Performance: System handles ~100 concurrent users okay; needs scaling for more
Challenges Faced and Resolved
- Challenge: Integrating map APIs for route optimization
Resolution: Used Google Maps API with distance matrix for optimized routes
Challenge: Handling delayed updates

Java Programming Project Presentation

Conclusion & Future Enhancement

The system effectively tracks parcels, updates statuses, and assigns drivers with optimized routes. It enhances transparency for customers and efficiency for logistics management. Future enhancements can further boost performance and user experience.

- Real-time Tracking with Live Maps: Show parcel location on a map
- Automated Notifications: More proactive alerts (e.g., estimated delivery time)
- Route Optimization AI: Improve routing with machine learning
- Customer Feedback Integration: Collect ratings for drivers/parcel handling
- Multi-language Support: Cater to more users with language options

Java Programming Project Presentation

References / Acknowledgement

- **Data Communications and Networking – Behrouz A. Forouzan**
- **Computer Networks – Andrew S. Tanenbaum**
- **MySQL Documentation**
- **Visual Studio Code Documentation**
- **GeeksforGeeks – <https://www.geeksforgeeks.org>**

I would like to express my sincere gratitude to my project guide and faculty members for their continuous support and guidance throughout the development of this project on Courier Parcel Tracking System.

I also thank my institution for providing the necessary facilities and resources to complete this project successfully.

I extend my heartfelt thanks to my friends and classmates for their valuable suggestions and support.

Finally, I would like to thank my family for their encouragement and motivation in completing this project.

Thank You

“Delivering Trust Through Technology”

Motto- `code.track.Deliver`